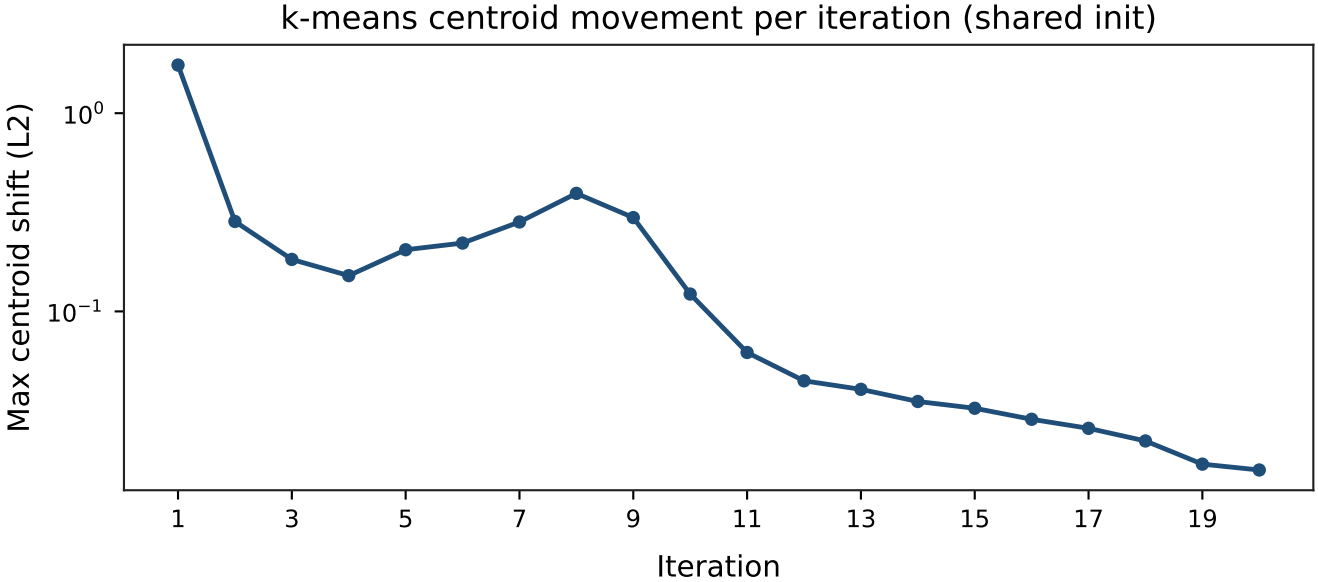

Spark and MapReduce used the same initial centroids and matching shift each iteration.

Neither reached convergence threshold $\varepsilon = 10^{-4}$ within 20 iterations.